TCP/IP & OSI Models — Study Notes

> 💡 **Big idea: Encapsulation** wraps data as it moves **down** the sender's protocol stack. **Decapsulation** unwraps it as it moves **up** the receiver's stack. Routers generally process up to Layer 3; they do **not** inspect TCP/UDP ports.

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1. Common 5-Layer TCP/IP Model

Layer	Main job	Examples	PDU / unit	Key note
5. Application	User-facing network services and protocols	HTTP/HTTPS, DNS, SMTP, FTP	Data / message	The interface between applications and the network
4. Transport	End-to-end, process-to-process delivery	TCP, UDP	**Segment** (TCP) / **Datagram** (UDP)	Not hop-by-hop; routers do not process TCP/UDP ports
3. Network / Internet	Logical addressing and routing	IPv4, IPv6, ICMP	**Packet**	Routers forward packets based on IP addresses
2. Data Link / Local Area	Local link delivery, MAC addressing, framing	Ethernet, Wi-Fi, PPP	**Frame**	Usually has a **header + trailer** (e.g., FCS for error detection)
1. Physical	Transmits raw bits/signals over media	Fiber, UTP, radio	**Bits**	No header/trailer at this layer

> ⚠️ **Terminology correction:** Your "Internet Layer" is commonly called the **Network Layer**. Your "Local Area Layer" is commonly called the **Data Link Layer** or **Network Access Layer**.

> 💡 **Layer support:** Lower layers provide services to the layers above them. For example, the Transport Layer relies on the Network Layer to move packets across networks.

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2. Encapsulation Flow — Sender Side

Encapsulation happens **on the sending host before transmission**.

```
Application data
   ↓ + Transport header
Segment (TCP) / Datagram (UDP)
   ↓ + Network/IP header
Packet
   ↓ + Data Link header + trailer
Frame
   ↓
Physical bits / signals
```

Example: Downloading a file over HTTPS

1. **Application:** HTTP/HTTPS request is created.
2. **Transport:** TCP adds a header → **TCP segment**.
3. **Network:** IP adds a header → **IP packet**.
4. **Data Link:** Ethernet/Wi-Fi adds header + trailer → **frame**.
5. **Physical:** Frame is transmitted as bits/signals.

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3. Decapsulation — Receiver Side

Decapsulation is the **reverse process** on the receiving host:

```
Physical bits/signals
   ↓
Frame
   ↓ remove Data Link header/trailer
Packet
   ↓ remove Network/IP header
Segment / Datagram
   ↓ remove Transport header
Application data
```

> 💡 At the physical layer, bits arrive **serially** at a given **bits-per-second (bps)** rate. Decapsulation is not a simple "left-to-right bit read"; the receiver reassembles bits into a frame, then parses structured headers layer by layer.

> 🔁 **Routers:** A router usually removes the incoming Layer 2 frame, inspects the Layer 3 IP packet, then re-encapsulates it in a new Layer 2 frame for the next hop. It does not normally decapsulate up to Layer 4.

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4. PDU Names Cheat Sheet

Layer	PDU name
Application	Data / Message
Transport	**Segment** (TCP) / **Datagram** (UDP)
Network	**Packet**
Data Link	**Frame**

Layer	PDU name
Physical	**Bits**

5. TCP/IP vs. OSI

> ⚠️ **Correction:** The **OSI model** is a **7-layer reference model** created by ISO. It is not a "revised 5-layer model." The TCP/IP model is the practical protocol suite used on the Internet.

OSI Layer	Role	Maps roughly to TCP/IP
7. Application	Network services for user applications	Application
6. Presentation	Data format, encoding, compression, encryption	Application
5. Session	Starts, manages, and ends sessions/dialogues	Application
4. Transport	End-to-end delivery, reliability, flow control	Transport
3. Network	Logical addressing and routing	Network / Internet
2. Data Link	Framing, MAC addressing, local-link delivery	Data Link
1. Physical	Bits and signals over physical media	Physical

> 💡 **Why OSI matters:** It gives a vendor-neutral reference model so devices and software from different companies can interoperate.

▼ 6. Common Pitfalls / Corrections

- ❌ "Transport Layer is hop-to-hop."
  ✅ **Transport is end-to-end (process-to-process). Data Link** is hop-to-hop.
- ❌ "ISO revised the 5 layers."
  ✅ ISO created the **7-layer OSI reference model**. TCP/IP is a separate practical model/suite.
- ❌ "Encapsulation happens during delivery."
  ✅ Encapsulation happens **on the sender before transmission**; decapsulation happens **on the receiver**.
- ❌ "Frame is just a header."
  ✅ A frame typically has a **header and a trailer** (e.g., FCS).
- ❌ "Decapsulation reads bits left-to-right."
  ✅ Bits arrive serially at the physical layer, then the receiver parses **structured headers layer by layer**.